Intermediate Elective 2

Minh Tri Ngo

Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# imported libraries from example
library(showtext)
library(camcorder)
library(ggtext)
library(glue)
library(ggfx)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))


# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")



# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
```

```r
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")


# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
```

```r
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪  = TRUE),
         site_full = fct_rev(site_full))

# creating new object from clean aquatic inverts
copepods_df <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# creating new clean object from aquatic inverts
ab_do <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
         # convert taxon to title
         taxon = to_any_case(taxon, case = "title"),
         # order taxon levels by abudance
         taxon = fct_reorder(taxon, ave_lit, .fun="max"))
```

1. Explain Inspiration

- The visualization I chose for my inspiration was the “Can an exploding snowman predict the summer season” by Nicola Rennie.
- I felt that this visualization makes sense as I would like to analyze temperature in relation to the different taxon availability and the graph allows me to analyze and communicate my findings the msot efficiently.
- The variables that I will choose in my dataset to show my visualization will be dissolved, taxon, and average abundance. In terms of visual components, the dissolved will be on

the x axis. Taxon will have different colors to differentiate. And the y axis will have the average abundance.

2. Plan figure

```r
knitr::include_graphics(here("data", "20260515_174533.jpg"))
```

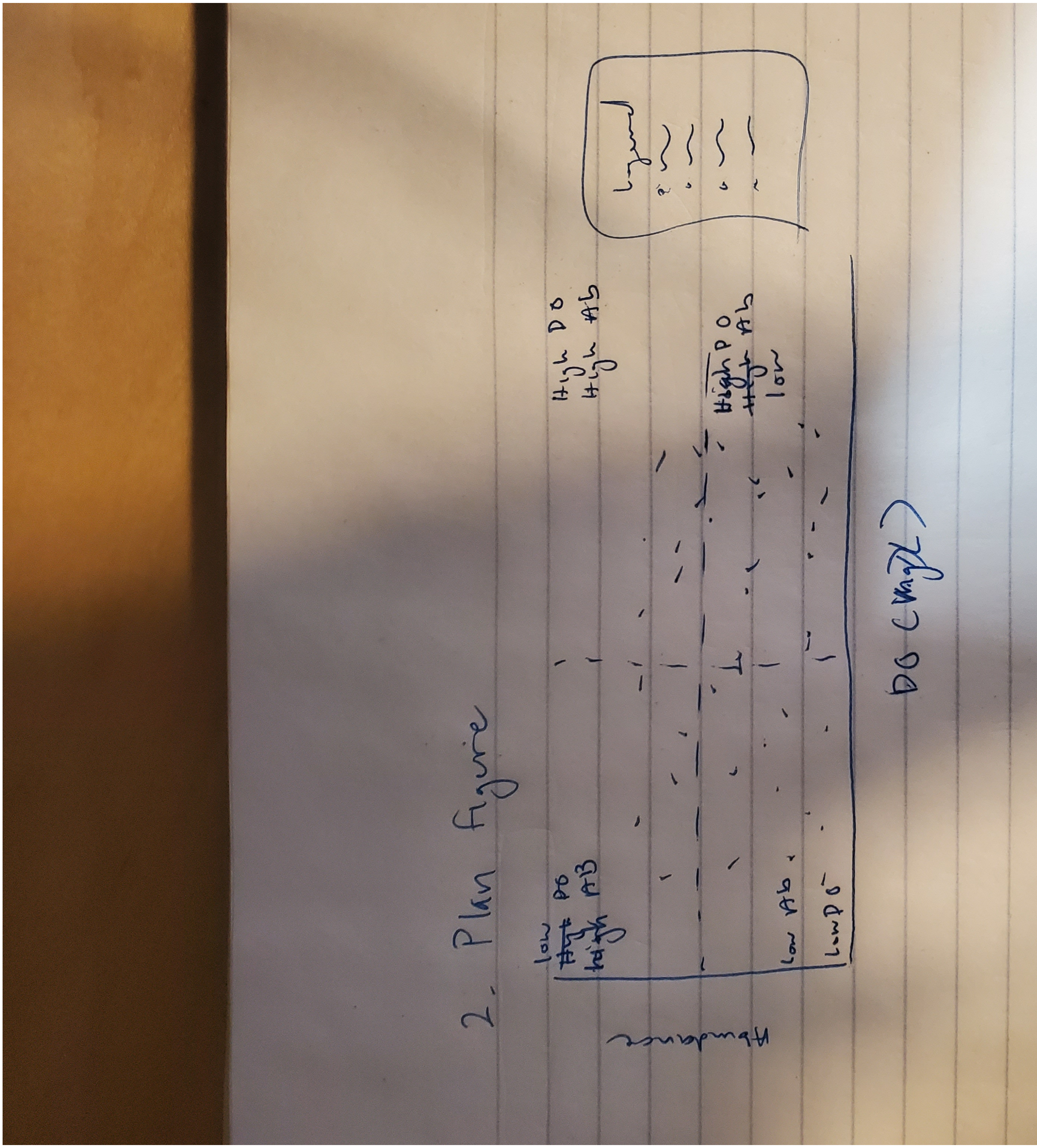


Figure 1: Draft photo

3. Code figure

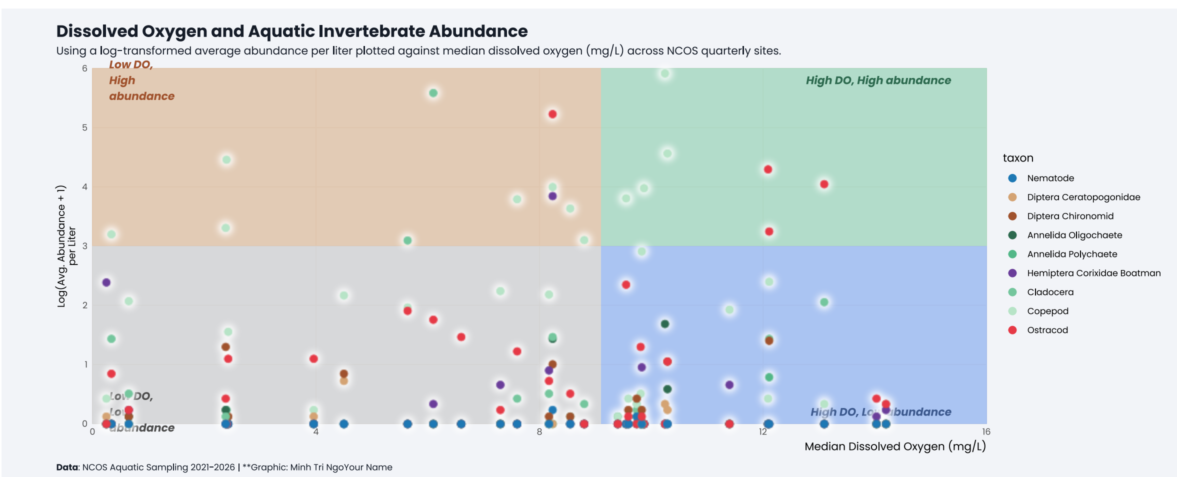
```r
# Load fonts ------------------------------------------------------------

font_add_google("Poppins", "Poppins")
showtext_auto()
showtext_opts(dpi = 300)
title_font <- "Poppins"
body_font <- "Poppins"


# Define colours and fonts--------------------------------------------------

bg_col <- "#F2F4F8"
text_col <- "#151C28"
highlight_col <- "cornflowerblue"

# Taxon color palette (one color per taxon)
taxon_colors <- c(
  "Annelida Oligochaete"          = "#2D6A4F",
  "Annelida Polychaete"           = "#52B788",
  "Cladocera"                     = "#74C69D",
  "Copepod"                       = "#B7E4C7",
  "Diptera Ceratopogonidae"       = "#D4A373",
  "Diptera Chironomid"            = "#A0522D",
  "Hemiptera Corixidae Boatman"   = "#6A3D9A",
  "Nematode"                      = "#1D78B4",
  "Ostracod"                      = "#E63946"
)


# Data prep --------------------------------------------------------------

plot_data <- ab_do |>
  filter(!is.na(med_do), !is.na(log_ave_lit))

med_do_val  <- median(plot_data$med_do,      na.rm = TRUE)
med_ab_val  <- 3


# Define text -------------------------------------------------------------
```

```r
social <- "Your Name"
icon <- "<span
 ↪ style='font-size:8pt;font-family:\"FA\";color:#3C598E;'>&#xf111;</span> "
title <- "Dissolved Oxygen and Aquatic Invertebrate Abundance"
st <- glue("Using a log-transformed average abundance per liter plotted
 ↪ against median dissolved oxygen (mg/L) across NCOS quarterly sites.")
cap <- paste0(
  "**Data**: NCOS Aquatic Sampling 2021-2026 | **Graphic: Minh Tri Ngo",
   ↪ social
)

most_abundant <- ab_do |>
  slice_max(log_ave_lit)

highest_do <- ab_do |>
  slice_max(med_do)

# Plot --------------------------------------------------------------------


ggplot(
  data = plot_data
) +
  annotate(
    "rect",
    xmin = med_do_val, xmax = Inf,
    ymin = med_ab_val, ymax = Inf, alpha = 0.5,
    fill = "#74C69D"
  ) +
  annotate(
    "text",
    x = 15.5, y = 5.8, label = "High DO, High abundance",
    colour = "#2D6A4F", family = body_font, fontface = "bold.italic",
    hjust = 1.05
  ) +
  annotate(
    "rect",
    xmin = med_do_val, xmax = Inf,
    ymin = -Inf, ymax = med_ab_val, alpha = 0.5,
    fill = highlight_col
  ) +
  annotate(
    "text",
```

```
    x = 15.5, y = 0.2, label = "High DO, Low abundance",
    colour = "#3C598E", family = body_font, fontface = "bold.italic",
    hjust = 1.05
  ) +
  annotate(
    "rect",
    xmin = -Inf, xmax = med_do_val,
    ymin = med_ab_val, ymax = Inf, alpha = 0.5,
    fill = "#D4A373"
  ) +
  annotate(
    "text",
    x = 0.3, y = 5.8, label = "Low DO,\nHigh\nabundance",
    colour = "#A0522D", family = body_font, fontface = "bold.italic",
    hjust = 0
  ) +
  annotate(
    "rect",
    xmin = -Inf, xmax = med_do_val,
    ymin = -Inf, ymax = med_ab_val, alpha = 0.5,
    fill = "grey"
  ) +
  annotate(
    "text",
    x = 0.3, y = 0.2, label = "Low DO,\nLow\nabundance",
    colour = "grey30", family = body_font, fontface = "bold.italic",
    hjust = 0
  ) +
  with_outer_glow(
    geom_point(
      mapping = aes(
        x = med_do, y = log_ave_lit,
        colour = taxon
      ),
      size = 3
    ),
    colour = "white",
    expand = 3
  ) +
  scale_colour_manual(values = c(
    "Annelida Oligochaete"        = "#2D6A4F",
    "Annelida Polychaete"         = "#52B788",
    "Cladocera"                   = "#74C69D",
```

```r
    "Copepod"                      = "#B7E4C7",
    "Diptera Ceratopogonidae"      = "#D4A373",
    "Diptera Chironomid"           = "#A0522D",
    "Hemiptera Corixidae Boatman" = "#6A3D9A",
    "Nematode"                     = "#1D78B4",
    "Ostracod"                     = "#E63946"
  )) +
  scale_x_continuous(limits = c(0, 16)) +
  scale_y_continuous(limits = c(0, 6)) +
  labs(
    x = "Median Dissolved Oxygen (mg/L)",
    y = "Log(Avg. Abundance + 1)\nper Liter",
    title = title, subtitle = st,
    caption = cap
  ) +
  coord_cartesian(expand = FALSE, clip = "off") +
  theme_minimal(base_size = 10, base_family = body_font) +
  theme(
    legend.position = "right",
    plot.margin = margin(10, 20, 10, 50),
    plot.title.position = "plot",
    plot.caption.position = "plot",
    plot.background = element_rect(fill = bg_col, colour = bg_col),
    panel.background = element_rect(fill = bg_col, colour = bg_col),
    plot.title = element_textbox_simple(
      colour = text_col,
      hjust = 0,
      halign = 0,
      margin = margin(b = 0, t = 5),
      family = title_font,
      face = "bold",
      size = rel(1.5)
    ),
    plot.subtitle = element_textbox_simple(
      colour = text_col,
      hjust = 0,
      halign = 0,
      margin = margin(b = 10, t = 5),
      family = body_font
    ),
    plot.caption = element_textbox_simple(
      colour = text_col,
      hjust = 0,
```

```
    halign = 0,
    margin = margin(b = 0, t = 10),
    family = body_font
  ),
  strip.text = element_textbox_simple(
    face = "bold",
    margin = margin(t = 10),
    size = rel(0.9)
  ),
  panel.grid.minor = element_blank(),
  axis.title.x = element_text(hjust = 1, margin = margin(t = 5)),
  axis.title.y.left = element_text(
    angle = 90,
    vjust = 0.5,
    family = body_font,
    size = rel(0.9),
    margin = margin(r = 5)
  )
)
```



4. Describe Process

- The process to use someone else's code to create a visualization make me creative in other aspects. While I had an image or preconcieved idea of what my graph would look like, I had to manipulate the code in my own way to be able to get the graphs to work.

- Challenges I encountered in this specific visualization adaptation would be that they didn't clarify the font packages were previously downloaded so I had to improvise. Additionally, there were some aspects of the graph that wouldn't work with my own as the median is 0 for some values, leading to the graph not creating 4 regions.
- This visualization differs from other assignments or electives since there is a higher emphasis on the visuals. Additionally, I split one graph into 4 sections, something that I hadn't done in other electives. I also was able to add quotes as well to cite data or to add context to the title.

```r
list.files("data")
```

```
character(0)
```